School of Informatics & Cybersecurity - Assessment Brief Coversheet

COMP H2037: Front End Development

Assessment Information

Module Title:	**Front End Development**
Module Code:	**COMP H2037**
Assessment Title:	**in-class lab test**
Assessment Type (e.g. Essay, Report, Presentation, Code submission):	**Coding in-class lab test**
Assessment Weighting (%):	**30%**
Submission Deadline:	**During your week 9 computer lab**
Submission mechanism:	**Brightspace - upload ZIP of folder: `/src`**
Late Submission Penalty:	**No late work will be accepted**

Use of AI

1	**NO AI**	The assessment is completed entirely without AI assistance. This level ensures that students rely solely on their own knowledge, understanding, and skills. **AI Must not be used at any point during the assessment.**	✓
2	**AI-ASSISTED IDEA GENERATION AND STRUCTURING**	AI can be used in the assessment for brainstorming, creating structures, and generating ideas for improving work. **No AI content is allowed in the final submission.**	☐
3	**AI-ASSISTED EDITING**	AI can be used to make improvements to the clarity or quality of student created work to improve the final output, but no new content can be created using AI. **AI can be used, but your original work with no AI content must be provided in an appendix.**	☐
4	**AI TASK COMPLETION, HUMAN EVALUATION**	AI is used to complete certain elements of the task, with students providing discussion or commentary on the AI-generated content. This level requires critical engagement with AI-generated content and evaluating its output. **You will use AI to complete specified tasks in your assessment. Any AI created content must be cited.**	☐
5	**FULL AI**	AI should be used as a 'co-pilot' to meet the requirements of the assessment, allowing for a collaborative approach with AI and enhancing creativity. **You may use AI throughout your assessment to support your own work and do not have to specify which content is AI-generated.**	☐

Further conditions for the use of AI not covered above:

No AI is permitted during this test
- You must turn off AI assistance in whatever editor you use to complete the test
- You may not consult any AI agent for help during the test, whether in a browser or otherwise

Academic term	**2025/26**
Year of study	**Year 2**
Semester	**2**
Time / location	**You have 95 minutes in total** **= 90 minutes for the test + 5 minutes to ZIP and upload**

COMP H2037: Front End Development

in-class lab test - SAMPLE

Internal Examiner(s)	**Dr. Matt Smith**

Instructions to candidates:

1.	**You are to create a new SvelteKit project using the command line tool.** **Then add features to this new project to meet the requirements of the test questions.**
2.	**This is an <u>individual</u> test – all work should be your own.** **During the test you should NOT communicate with anyone apart from the lecturer (excepting medical / emergency circumstances).**
3.	**NO AI is permitted during this test**
4.	**NO communication with others is permitted** **(e.g. to NOT use phones, email etc.)**
5.	**NO watching of videos or use of headphones/earphones is permitted**
6.	**You MUST use a university PC or your own laptop for this test.** *NOTE: days/hours before the test ensure you know your password and can login!*
7.	**This is an open-book test for paper-based & electronic materials.** - **You may look up paper and online examples/code reference during the test, as long as no AI help is used**
8.	**It is recommended that you answer the questions <u>in the sequence they appear</u>, since some questions rely on your work created in earlier questions.**
9.	**Answer ALL questions. The test is marked out of 100.**

UPLOAD TO BRIGHTSPACE ONE ZIP folder from your project (NOT the whole project)

- **`src.zip`** – a ZIP of your **`/src`** folder

Question 1 – Routes – Logic 10 marks

(a) all pages

- the HTML titles of each page should be route name between spaces and two dashes
  - e.g. **`-- home --, -- privacy --`**
- every page should have a nav bar with links to:
  - **`home`**
  - **`privacy`**

    NOTE: Ensure the link text is the same as the route name

(b) route: /

- route name: **home**
- Body content
  - level 1 heading "**Welcome to our website**"
  - level 2 heading "**have a nice day**"

(c) route: **`/privacy`**

- route name: **privacy**
- Body content:
  - level 1 heading "**Privacy Policy**"
  - paragraph: "**we collect as much data as we can to improve our marketing emails**"

Question 2 – Utility functions 20 marks

Create file **`/src/lib/util/useful_functions.js`**

(a) In this file declare and export function: **`isEven(n)`**

- It takes in one parameter: **`n`**
  - An integer (whole number)
- the function should return values according to the following logic:
  - IF **`n`** is even THEN return **true**
  - Otherwise the function should return **false**

(b) In this file declare and export function: **`isNeutral(ph)`**

- It takes in one parameter: **`ph`**
  - a decimal (real) number representing a Ph acidity value
- the function should return values according to the following logic:
  - IF **`ph`** is exactly 7 THEN return **true**
  - Otherwise the function should return **false**

Question 3 – Route with data 20 marks

(a) add another route: **`/oddeven/<n>`**

- route name: **oddeven**
- gets data from URL
- Body content
  - paragraph showing message about whether number is ODD or EVEN as illustrated:

    URL: **/oddeven/9** >> **9 is an ODD number**

    URL: **/oddeven/4** >> **4 is an EVEN number**
- You must write code to make the page output the appropriate message for whatever integer is provided at the end of the URL
  - do NOT hard-code the script for only certain numbers
  - Use one of the functions defined in Question 2

(b) if no number is provided (**`/oddeven`**)

- then output the following error message
  - level 1 heading "**Error – missing parameter**"
  - paragraph:
    "**you have to provide a number at the end of the URL, e.g. `/oddeven/6`**"

Question 4 page created objects and loop to display 20 marks

(a) Add a new page to your site to display objects for the entity class you created in the previous question.

- **route: `/food`**
- the page script should create a JSON array of the following 3 objects, representing food names and their Ph values:

lemon juice	2
lettuce	7
apples	4

The page should loop through the array to display the objects in the form of an **HTML table** as show above.

(b) Add logic to your page so that a third column is added to the table as shown below

- use one of the functions defined in Question 2 to test each object's Ph value, to determine which text to display in this third column:

lemon juice	2	Acidic or alkaline
lettuce	7	Neutral
apples	4	Acidic or alkaline

(c) add a link to link to this route to the global site navigation bar

Question 5 – JSON data file 10 marks

(a) Rather than the JSON being created as a variable in the pager script, instead declare the data in file **`/src/lib/data/foods.json`**

And modify your page script to read the data from this JSON file.

Question 6 – Form sends data 20 marks

(a) add another route: **`/evenform`**

- route name: **evenform**
- this page presents a form to the user, asking them to enter a number
  - (using the POST form submission method)

(b) add a link to link to this route to the global site navigation bar

(c) Write the JavaScript code to process the form submission

- And pass the received number back to the **`/evenform`** Svelte page

(d) If the Svelte page receives a number from a form submission, it should output a message showing message about whether number is ODD or EVEN as illustrated, e.g.:

3 is an ODD number

22 is an EVEN number

(e) Also make the JavaScript code that processes the form submission write the number received into a cookie named 'number'

HINT:

- the load() function should retrieve the cookie value and pass it as data to the Svelte page
- the Svelte page should
  - always display the form
  - only display the message about the even'ness of the number received from the form if such a form value is received